University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2022

Answer all questions.
Approximate marks per question are shown in brackets.

Each section MUST be written in a separate answer booklet.

Total Questions	**:**	**5 questions**
Total Pages (including cover)	**:**	**4 pages**

Time: 45 minutes **Marks: 40**

Section 1. Project Management [26 Marks]

Question 1 [6 Marks]

a. Consider the following values in the Agile Manifesto: [6]

A. Working software

B. Responding to change

For each, give the "item on the right" that they are contrasted with (e.g. customer collaboration over contract negotiation), give an example of an agile principle which captures some aspect of the value, and explain the value and its importance in 1-2 sentences.

ANSWER:

1 point for the correct item on the right

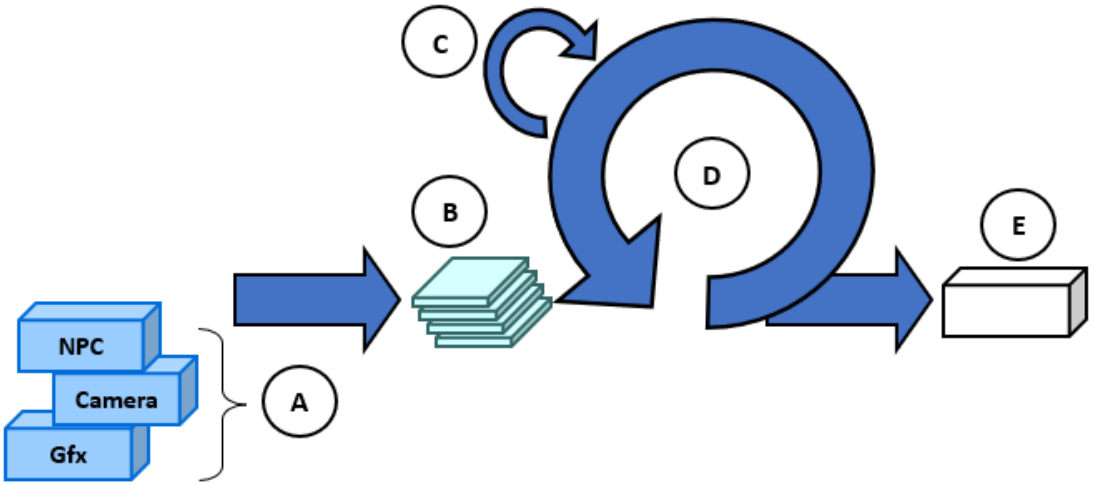
1 point for naming an appropriate agile principle

1 point for clearly explaining the value and why it is important, it has to do more than restate the value

Working software over comprehensive documentation: Prioritize building working software early over doing the design all at the beginning; this means early versions will not represent all possible requirements, but will represent the highest priority items. It is important because it means that a deliverable version of the software is delivered earlier in the project; with lower priority items added as allowed by time and cost

Responding to change over following a plan: Expect the system requirements to change and so design the system to accommodate these changes

Question 2 [10 Marks]



For each item labelled above (A-E), explain what it is, who is responsible for managing or evaluating it, and any associated time-frames. 2 marks per item. [10]

ANSWER:

½ point for the name, ½ point for explanation, ½ for the manager or evaluator, ½ for the time frame.

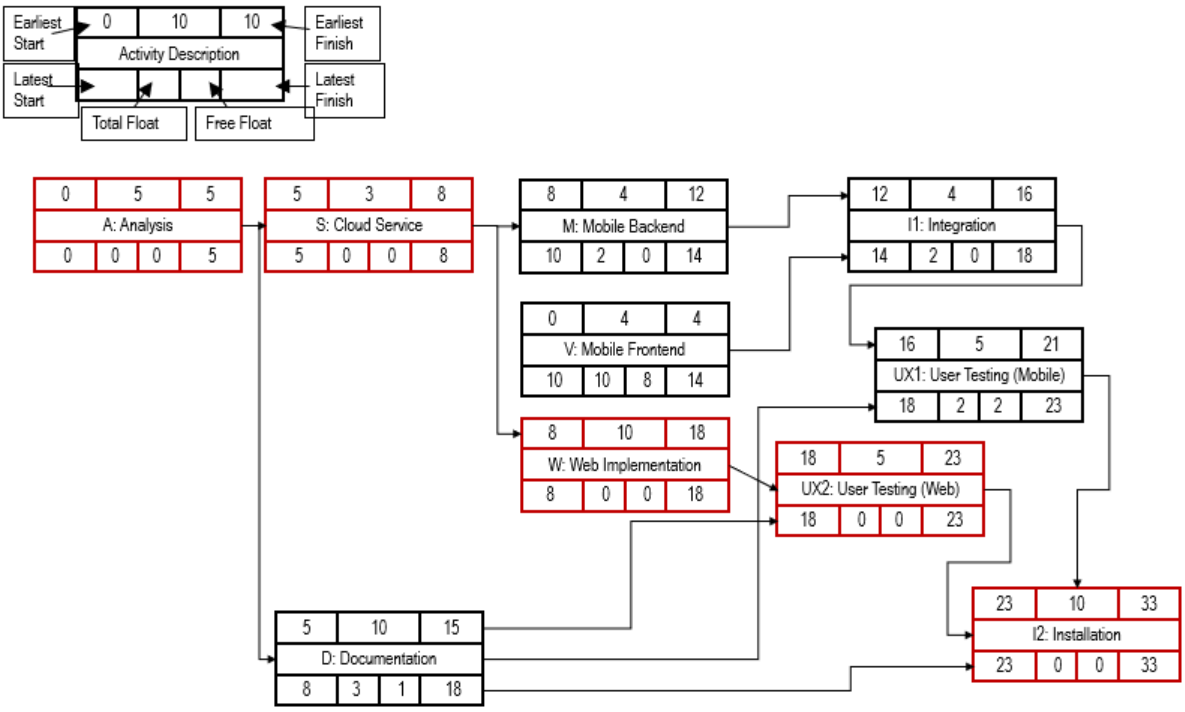
A. Product Backlog – list of tasks to be completed for the project
    a. Managed by Product Owner
    b. Endures across the project (multiple sprints)
B. Sprint Backlog or Backlog Tasks – list of tasks to be completed for the print
    a. Managed by the Team
    b. Endures for a singles sprint
C. Daily Scrum Meeting – a short update meeting each day to help team synchronize and identify issues
    a. Managed by the Scrum Master
    b. Daily – usually 15 minutes at the start of each day
D. Sprint – a period of time in which a subset of the backlog is implemented and tested for delivery
    a. Managed by the Team
    b. Usually 30 days, but could be 2-6 weeks depending on the time (answer just needs to say 30 days but the range is acceptable.)
E. Product Increments – a version of the product that should be potentially shippable (i.e. tested, packaged)
    a. Managed by the Team; Evaluated by the Customer
    b. Delivered at the end of every sprint – so every 30 days.

Question 3 [10 Marks]

a. Draw a network analysis diagram based on the following task list and identify the critical path. [10]

Task	Time	Dependencies
A: Analysis and Architecture; API definitions	5	
S: Cloud Service Implementation	3	A
M: Mobile App Backend Implementation	4	S
V: Mobile App Frontend Implementation	4	
W: Web Implementation	10	S
I1: Integration of Mobile App	4	M, V
UX1: User Testing – Mobile App	5	I1, D
UX2: User Testing – Web App	5	W, D
D: Documentation	10	A
I2: Installation and Customer Hand Over	10	D, UX1, UX2

ANSWER:



[3] Dependencies: 3 points total, minus half point for each error

[2] Critical Path: 2 points for correct identification of critical path, 1 point for partially correct

Check the following numbers

[3] Earliest Finish: 1 point for getting I2: Installation correct, 1 for I1: Integration, 1 for UX2

[1] Total float: Latest Start – Earliest Start (1 point for correct calculations, even if the numbers for LS and ES are incorrect)

[1] Free float: ES (next) – EF (current) (l point for correct calculations)

Section 2. Architecture [14 Marks]

Question 4 [6 Marks]

a. What is Software Quality about? [2]

ANSWER:

SQ is about the study and practice that defines the desirable attributes of software products.

b. State any four runtime Software Quality. [4]

ANSWER:

(i) Performance, (ii) Security, (iii) Availability, (iv) Functionality, (v) Usability

Question 5 [8 Marks]

c. Briefly explain Inheritance and Polymorphism. [4]

ANSWER:

(i) Inheritance: Entities can inherit attributes from other entities. That is parent classes can share/extend their behaviours and attributes with/to child classes. This also means that reusability is possible [2].

(ii) Polymorphism: Entities can have two or more forms. That is objects are designed (or can be designed) to share behaviors. By overloading or overriding, the same method can perform different behaviors [2].

d. What is UML used for with respect to software artifacts? [4]

ANSWER:

(i) Visualizing, (ii) Specifying, (iii) Constructing, (iv) Documenting

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